

```
import pandas as pd
import numpy as np
```

```
data = pd.read_csv('./Football.csv')
data
```

Country		League	Club	Player Names				
Matches_Played \								
0	Spain	La Liga	(BET)	Juanmi Callejon				
19								
1	Spain	La Liga	(BAR)	Antoine Griezmann				
36								
2	Spain	La Liga	(ATL)	Luis Suarez				
34								
3	Spain	La Liga	(CAR)	Ruben Castro				
32								
4	Spain	La Liga	(VAL)	Kevin Gameiro				
21								
..	...	...	...	...				
.								..
655	Netherlands	Eredivisie	(UTR)	Gyrano Kerk				
24								
656	Netherlands	Eredivisie	(AJA)	Quincy Promes				
18								
657	Netherlands	Eredivisie	(PSV)	Denzel Dumfries				
25								
658	Netherlands	Eredivisie	NaN	Cyriel Dessers				
26								
659	Netherlands	Eredivisie	(PSV)	Cody Gakpo				
14								
Substitution		Mins	Goals	xG	xG Per	Avg Match	Shots	
OnTarget \								
0	16	1849	11	6.62		0.34	48	
20								
1	0	3129	16	11.86		0.36	88	
41								
2	1	2940	28	23.21		0.75	120	
57								
3	3	2842	13	14.06		0.47	117	
42								
4	10	1745	13	10.65		0.58	50	
23								
..	...	...	...	...		...	...	
...								
655	0	2155	10	7.49		0.33	50	
18								
656	2	1573	12	9.77		0.59	56	
30								

657	0	2363	7	5.72	0.23	45
14						
658	0	2461	15	14.51	0.56	84
43						
659	11	1557	7	4.43	0.27	38
15						

	Shots Per	Avg Match	On Target Per	Avg Match	Year
0		2.47		1.03	2016
1		2.67		1.24	2016
2		3.88		1.84	2016
3		3.91		1.40	2016
4		2.72		1.25	2016
..		...		...	...
655		2.20		0.79	2020
656		3.38		1.81	2020
657		1.81		0.56	2020
658		3.24		1.66	2020
659		2.32		0.92	2020

[660 rows x 15 columns]

Reading csv from github

```
link = 'https://raw.githubusercontent.com/AshishJangra27/Data-Analysis-with-Python-GFG/main/3.%20Data%20Preprocessing%20-%20Removing%20Null%20Value%20Rows/googleplaystore.csv'
```

```
df = pd.read_csv(link) #This will fetch the data from the internet
df
```

	App
Category \	
0	Photo Editor & Candy Camera & Grid & ScrapBook
ART_AND_DESIGN	
1	Coloring book moana
ART_AND_DESIGN	
2	U Launcher Lite – FREE Live Cool Themes, Hide ...
ART_AND_DESIGN	
3	Sketch - Draw & Paint
ART_AND_DESIGN	
4	Pixel Draw - Number Art Coloring Book
ART_AND_DESIGN	
...	...
...	
10836	Sya9a Maroc - FR
FAMILY	
10837	Fr. Mike Schmitz Audio Teachings
FAMILY	
10838	Parkinson Exercices FR

MEDICAL

10839

The SCP Foundation DB fr nn5n

BOOKS_AND_REFERENCE

10840

iHoroscope - 2018 Daily Horoscope & Astrology

LIFESTYLE

	Rating	Reviews	Size	Installs	Type	Price	\
0	4.1	159	19M	10,000+	Free	0	
1	3.9	967	14M	500,000+	Free	0	
2	4.7	87510	8.7M	5,000,000+	Free	0	
3	4.5	215644	25M	50,000,000+	Free	0	
4	4.3	967	2.8M	100,000+	Free	0	
...	...	...	...	...	...	...	
10836	4.5	38	53M	5,000+	Free	0	
10837	5.0	4	3.6M	100+	Free	0	
10838	NaN	3	9.5M	1,000+	Free	0	
10839	4.5	114	Varies with device	1,000+	Free	0	
10840	4.5	398307	19M	10,000,000+	Free	0	

	Content Rating	Genres	Last Updated	\
0	Everyone	Art & Design	January 7, 2018	
1	Everyone	Art & Design;Pretend Play	January 15, 2018	
2	Everyone	Art & Design	August 1, 2018	
3	Teen	Art & Design	June 8, 2018	
4	Everyone	Art & Design;Creativity	June 20, 2018	
...	...	...	...	
10836	Everyone	Education	July 25, 2017	
10837	Everyone	Education	July 6, 2018	
10838	Everyone	Medical	January 20, 2017	
10839	Mature 17+	Books & Reference	January 19, 2015	
10840	Everyone	Lifestyle	July 25, 2018	

	Current Ver	Android Ver
0	1.0.0	4.0.3 and up
1	2.0.0	4.0.3 and up
2	1.2.4	4.0.3 and up
3	Varies with device	4.2 and up
4	1.1	4.4 and up
...	...	...
10836	1.48	4.1 and up
10837	1.0	4.1 and up
10838	1.0	2.2 and up
10839	Varies with device	Varies with device
10840	Varies with device	Varies with device

[10841 rows x 13 columns]

df = data

df.info() *# Gives the basic info of the dataset*

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 660 entries, 0 to 659
Data columns (total 15 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Country                               660 non-null    object
1   League                                660 non-null    object
2   Club                                  626 non-null    object
3   Player Names                          660 non-null    object
4   Matches_Played                        660 non-null    int64
5   Substitution                          660 non-null    int64
6   Mins                                  660 non-null    int64
7   Goals                                 660 non-null    int64
8   xG                                    660 non-null    float64
9   xG Per Avg Match                     660 non-null    float64
10  Shots                                660 non-null    int64
11  OnTarget                             660 non-null    int64
12  Shots Per Avg Match                   660 non-null    float64
13  On Target Per Avg Match               660 non-null    float64
14  Year                                  660 non-null    int64
dtypes: float64(4), int64(7), object(4)
memory usage: 77.5+ KB

```

Cleaning the data using Pandas

#Checking null values

```

df.isnull() #gives true or false for the null values
df.isna()   #gives true or false for the null values

```

	Country	League	Club	Player Names	Matches_Played
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
655	False	False	False	False	False
656	False	False	False	False	False

```

657     False     False     False           False           False
False
658     False     False     True           False           False
False
659     False     False     False           False           False
False

      Mins  Goals      xG  xG Per Avg Match  Shots  OnTarget  \
0      False  False  False           False  False  False
1      False  False  False           False  False  False
2      False  False  False           False  False  False
3      False  False  False           False  False  False
4      False  False  False           False  False  False
..      ...    ...    ...           ...    ...    ...
655     False  False  False           False  False  False
656     False  False  False           False  False  False
657     False  False  False           False  False  False
658     False  False  False           False  False  False
659     False  False  False           False  False  False

      Shots Per Avg Match  On Target Per Avg Match  Year
0           False           False  False
1           False           False  False
2           False           False  False
3           False           False  False
4           False           False  False
..           ...           ...    ...
655           False           False  False
656           False           False  False
657           False           False  False
658           False           False  False
659           False           False  False

[660 rows x 15 columns]

```

Checking null values in each of the columns

```

df.isnull().sum()    #gives the number of null values in each of the
column

Country              0
League               0
Club                 34
Player Names         0
Matches_Played       0
Substitution         0
Mins                 0
Goals                0
xG                   0
xG Per Avg Match     0

```

```
Shots          0
OnTarget       0
Shots Per Avg Match  0
On Target Per Avg Match  0
Year           0
dtype: int64
```

Quantile function

`df.describe()` *#gives the basic statistical calculation of the dataset*

	Matches_Played	Substitution	Mins	Goals
xG \				
count	660.000000	660.000000	660.000000	660.000000
mean	22.371212	3.224242	2071.416667	11.810606
std	9.754658	3.839498	900.595049	6.075315
min	2.000000	0.000000	264.000000	2.000000
25%	14.000000	0.000000	1363.500000	8.000000
50%	24.000000	2.000000	2245.500000	11.000000
75%	31.000000	5.000000	2822.000000	14.000000
max	38.000000	26.000000	4177.000000	42.000000

	xG Per Avg Match	Shots	OnTarget	Shots Per Avg
Match \				
count	660.000000	660.000000	660.000000	660.000000
mean	0.476167	64.177273	28.365152	2.948015
std	0.192831	34.941622	16.363149	0.914906
min	0.070000	5.000000	2.000000	0.800000
25%	0.340000	37.750000	17.000000	2.335000
50%	0.435000	62.000000	26.000000	2.845000
75%	0.570000	86.000000	37.000000	3.382500
max	1.350000	208.000000	102.000000	7.200000

	On Target Per Avg Match	Year
count	660.000000	660.000000
mean	1.315652	2018.363636
std	0.474239	1.367700
min	0.240000	2016.000000
25%	0.980000	2017.000000
50%	1.250000	2019.000000
75%	1.540000	2019.000000
max	3.630000	2020.000000

`df.describe(percentiles=[.80])` *#Will help in calculating custom percentiles*

	Matches_Played	Substitution	Mins	Goals
xG \				
count	660.000000	660.000000	660.000000	660.000000
660.000000				
mean	22.371212	3.224242	2071.416667	11.810606
10.089606				
std	9.754658	3.839498	900.595049	6.075315
5.724844				
min	2.000000	0.000000	264.000000	2.000000
0.710000				
50%	24.000000	2.000000	2245.500000	11.000000
9.285000				
80%	32.000000	6.000000	2915.800000	15.000000
14.076000				
max	38.000000	26.000000	4177.000000	42.000000
32.540000				

	xG Per Avg Match	Shots	OnTarget	Shots Per Avg
Match \				
count	660.000000	660.000000	660.000000	660.000000
mean	0.476167	64.177273	28.365152	2.948015
std	0.192831	34.941622	16.363149	0.914906
min	0.070000	5.000000	2.000000	0.800000
50%	0.435000	62.000000	26.000000	2.845000
80%	0.610000	90.000000	39.000000	3.600000
max	1.350000	208.000000	102.000000	7.200000

	On Target Per Avg Match	Year
count	660.000000	660.000000
mean	1.315652	2018.363636
std	0.474239	1.367700

```
min          0.240000  2016.000000
50%          1.250000  2019.000000
80%          1.630000  2020.000000
max          3.630000  2020.000000
```

```
df['Mins'].quantile(.80)    #We can calculate quantiles explicitly
2915.8
df['Mins'].quantile(.90)
3147.1
```

Copying a dataframe

```
newDf = data.copy() #This will not copy the reference but the actual
values
newDf
```

	Country	League	Club	Player Names
Matches_Played \				
0	Spain	La Liga	(BET)	Juanmi Callejon
19				
1	Spain	La Liga	(BAR)	Antoine Griezmann
36				
2	Spain	La Liga	(ATL)	Luis Suarez
34				
3	Spain	La Liga	(CAR)	Ruben Castro
32				
4	Spain	La Liga	(VAL)	Kevin Gameiro
21				
..	...	...	...	...
.				
655	Netherlands	Eredivisie	(UTR)	Gyrano Kerk
24				
656	Netherlands	Eredivisie	(AJA)	Quincy Promes
18				
657	Netherlands	Eredivisie	(PSV)	Denzel Dumfries
25				
658	Netherlands	Eredivisie	NaN	Cyriel Dessers
26				
659	Netherlands	Eredivisie	(PSV)	Cody Gakpo
14				

	Substitution	Mins	Goals	xG	xG Per	Avg Match	Shots
OnTarget \							
0	16	1849	11	6.62		0.34	48
20							
1	0	3129	16	11.86		0.36	88
41							
2	1	2940	28	23.21		0.75	120

57							
3	3	2842	13	14.06	0.47	117	
42							
4	10	1745	13	10.65	0.58	50	
23							
..	...	...	...	...	...	...	
...							
655	0	2155	10	7.49	0.33	50	
18							
656	2	1573	12	9.77	0.59	56	
30							
657	0	2363	7	5.72	0.23	45	
14							
658	0	2461	15	14.51	0.56	84	
43							
659	11	1557	7	4.43	0.27	38	
15							

	Shots Per Avg Match	On Target Per Avg Match	Year
0	2.47	1.03	2016
1	2.67	1.24	2016
2	3.88	1.84	2016
3	3.91	1.40	2016
4	2.72	1.25	2016
..	...	...	...
655	2.20	0.79	2020
656	3.38	1.81	2020
657	1.81	0.56	2020
658	3.24	1.66	2020
659	2.32	0.92	2020

[660 rows x 15 columns]

```
newDf["Year + 100"] = newDf['Year']+100 #Adding new column in the dataframe
newDf
```

	Country	League	Club	Player Names
Matches_Played \				
0	Spain	La Liga	(BET)	Juanmi Callejon
19				
1	Spain	La Liga	(BAR)	Antoine Griezmann
36				
2	Spain	La Liga	(ATL)	Luis Suarez
34				
3	Spain	La Liga	(CAR)	Ruben Castro
32				
4	Spain	La Liga	(VAL)	Kevin Gameiro
21				
..	...	...	...	...

655	Netherlands	Eredivisie	(UTR)	Gyrano Kerk
24				
656	Netherlands	Eredivisie	(AJA)	Quincy Promes
18				
657	Netherlands	Eredivisie	(PSV)	Denzel Dumfries
25				
658	Netherlands	Eredivisie	NaN	Cyriel Dessers
26				
659	Netherlands	Eredivisie	(PSV)	Cody Gakpo
14				

	Substitution	Mins	Goals	xG	xG Per	Avg Match	Shots
OnTarget \							
0	16	1849	11	6.62		0.34	48
20							
1	0	3129	16	11.86		0.36	88
41							
2	1	2940	28	23.21		0.75	120
57							
3	3	2842	13	14.06		0.47	117
42							
4	10	1745	13	10.65		0.58	50
23							
..	...	...	...	...		...	...
...							
655	0	2155	10	7.49		0.33	50
18							
656	2	1573	12	9.77		0.59	56
30							
657	0	2363	7	5.72		0.23	45
14							
658	0	2461	15	14.51		0.56	84
43							
659	11	1557	7	4.43		0.27	38
15							

	Shots Per	Avg Match	On Target Per	Avg Match	Year	Year + 100
0		2.47		1.03	2016	2116
1		2.67		1.24	2016	2116
2		3.88		1.84	2016	2116
3		3.91		1.40	2016	2116
4		2.72		1.25	2016	2116
..		...		...	...	...
655		2.20		0.79	2020	2120
656		3.38		1.81	2020	2120
657		1.81		0.56	2020	2120
658		3.24		1.66	2020	2120
659		2.32		0.92	2020	2120

```
[660 rows x 16 columns]
```

Value count functions

```
check = df['Player Names'].value_counts()    #Counts the repetitions of each values and sort it
print(check)
print(type(check))

Player Names
Andrea Belotti      5
Lionel Messi        5
Luis Suarez         5
Andrej Kramaric     5
Ciro Immobile       5
..
Francois Kamano    1
Lebo Mothiba        1
Gaetan Laborde      1
Falcao              1
Cody Gakpo          1
Name: count, Length: 444, dtype: int64
<class 'pandas.core.series.Series'>

check = df['Player Names'].unique() #This will give us the list of all the unique values in the column
check

array(['Juanmi Callejon', 'Antoine Griezmann', 'Luis Suarez',
      'Ruben Castro', 'Kevin Gameiro', 'Cristiano Ronaldo',
      'Karim Benzema', 'Neymar ', 'Iago Aspas', 'Sergi Enrich',
      'Aduriz ', 'Sandro Ramirez', 'Lionel Messi', 'Gerard Moreno',
      'Morata', 'Wissam Ben Yedder', 'Willian Jose', 'Andone ',
      'Cedric Bakambu', 'Isco', 'Mohamed Salah', 'Gregoire Defrel',
      'Ciro Immobile', 'Nikola Kalinic', 'Dries Mertens',
      'Alejandro Gomez', 'Jose Callejon', 'Iago Falque',
      'Giovanni Simeone', 'Mauro Icardi', 'Diego Falcinelli',
      'Cyril Thereau', 'Edin Dzeko', 'Lorenzo Insigne',
      'Fabio Quagliarella', 'Borriello ', 'Carlos Bacca',
      'Gonzalo Higuain', 'Keita Balde', 'Andrea Belotti', 'Fin
Bartels',
      'Lars Stindl', 'Serge Gnabry', 'Wagner ', 'Andrej Kramaric',
      'Florian Niederlechner', 'Robert Lewandowski', 'Emil Forsberg',
      'Timo Werner', 'Nils Petersen', 'Vedad Ibisevic', 'Mario
Gomez',
      'Maximilian Philipp', 'A\x8ldam Szalai',
      'Pierre-Emerick Aubameyang', 'Guido Burgstaller', 'Max Kruse',
      'Chicharito ', 'Anthony Modeste', 'Arjen Robben', 'Alexis
Sanchez',
```

'Romelu Lukaku', 'Harry Kane', 'Jamie Vardy', 'Christian Benteke',
 'Pedro None', 'Eden Hazard', 'Roberto Firmino', 'Sadio Mane',
 'Philippe Coutinho', 'Diego Costa', 'Dele Alli', 'Sergio Agüero',
 'Jermain Defoe', 'Fernando Llorente', 'Michail Antonio',
 'Zlatan Ibrahimovic', 'Olivier Giroud', 'Son Heung-Min',
 'Joshua King', 'Diego Souza', 'Pablo ', 'Robinho ', 'Kempes ',
 'Gabriel Jesus', 'Bruno Rangel ', 'Rogerio ', 'Vitor Bueno',
 'Marinho ', 'Grafite ', 'Andres Chavez', 'Cicero Semedo',
 'Sissa',
 'Giorgian de Arrascaeta', 'Keno ', 'Fred ', 'Kleber Gladiador',
 'Pottker ', 'Jonathan Copete', 'Ricardo Oliveira',
 'Angel Rodriguez', 'Gareth Bale', 'Rodrigo None', 'Sergio Leon',
 'Maxi Gomez', 'Mikel Oyarzabal', 'Willian Josa', 'Simone Zaza',
 'Portu ', 'Cristhian Stuani', 'Santi Mina', 'Morales ', 'Munir ',
 'Jose Callejon', 'Sergej Milinkovic-Savic', 'Duvan Zapata',
 'Paulo Dybala', 'Mirco Antenucci', 'Luis Alberto',
 'Roberto Inglese', 'Josip Ilcic', 'Ivan Perisic',
 'Leonardo Pavoletti', 'Kevin Lasagna', 'Niclas Fullkrug',
 'Salomon Kalou', 'Thorgan Hazard', 'Jean-Kevin Augustin',
 'Sandro Wagner ', 'Leon Bailey', 'Daniel Didavi',
 'Alfred Finnbogason', 'Davie Selke', 'Mark Uth',
 'Michael Gregoritsch', 'Julian Brandt', 'Kevin Volland', 'Roger ',
 'Arthur Caike', 'Everton ', 'Hernanes ', 'Luiz Fernando',
 'Wellington Paulista', 'Santiago Trellez', 'Jo', 'Thiago Neves',
 'Bruno Henrique', 'Dudu ', 'Diego ', 'Lucca ', 'Henrique Dourado',
 'Andre', 'Edigar Junio ', 'Junior Dutra', 'Jaime Mata',
 'Inaki Williams', 'Chimy Avila', 'Raul de Tomas', 'Pablo Sarabia',
 'Borja Iglesias', 'Jorge Molina', 'Charles', 'Arkadiusz Milik',
 'Mandzukic None', 'Andrea Petagna', 'Francesco Caputo',
 'Stephan El Shaarawy', 'Gervinho ', 'Krzysztof Piatek',
 'Alassane Plea', 'Kai Havertz', 'Luka Jovic', 'Ante Rebic',
 'Jadon Sancho', 'Ondrej Duda', 'Paco Alcacer', 'Benito Raman',
 'Wout Weghorst', 'Ishak Belfodil', 'Marco Reus',
 'Jean-Philippe Mateta', 'Sebastien Haller', 'Yussuf Poulsen',
 'Angel Di Maria', 'Remi Oudin', 'Nicolas Pepe', 'Emiliano Sala',
 'Jonathan Bamba', 'M'Baye Niang', 'Edinson Cavani',
 'Stephane Bahoken', 'Max Gradel', 'Florian Thauvin',
 'Kylian Mbappe-Lottin', 'Wahbi Khazri', 'Falcao ',
 'Gaetan Laborde', 'Andy Delort', 'Moussa Dembele', 'Memphis Depay',

'Lebo Mothiba', 'Francois Kamano', 'Luka Milivojevic',
 'Paul Pogba', 'Ashley Barnes', 'Glenn Murray', 'Richarlison ',
 'Callum Wilson', 'Gylfi Sigurdsson', 'Raheem Sterling',
 'Ayoze Perez', 'Alexandre Lacazette', 'Raul Jimenez',
 'Lucas Paqueta', 'Juan Cazares', 'Deyverson ', 'Leandro Damiao
 ',
 'Yago Pikachu', 'Rodrygo ', 'Andres Rios', 'Roger Guedes ',
 'Leandro Pereira ', 'Pedro ', 'Nico Lupez', 'Nene ', 'Gilberto
 ',
 'Henrique ', 'Willian ', 'Gabriel Barbosa', 'Diego Rossi',
 'Josef Martinez', 'Chris Wondolowski', 'Nani ', 'Kacper
 Przybylko',
 'Kei Kamara', 'C.J. Sapong', 'Gyasi Zardes', 'Heber ',
 'Mauro Manotas', 'Brian Fernandez', 'Alejandro Pozuelo',
 'Felipe Gutierrez', 'Jordan Morris', 'Raul Ruidiaz',
 'Jozy Altidore', 'Carlos Vela', 'Nemanja Nikolic',
 'Alexandru Mitrita', 'Joselu', 'Carlos Fernandez', 'Ante
 Budimir',
 'Lucas Perez', 'Loren Moron', 'Raul Garcla', 'Morata ',
 'Sergio Ramos', 'Lucas Ocampos', 'Cazorla ', 'Pote ',
 'Tiquinho Soares', 'Eduardo Mancha', 'Paulinho ', 'Alex
 Telles',
 'Bruno Viana', 'Mehrdad Mohammadi', 'Carlos Valenzuela',
 'Ruben Lameiras', 'Moussa Marega', 'Gian-Luca Waldschmidt',
 'Samuel Lino', 'Andre Andre', 'Mehdi Taremi', 'Carlos
 Vinicius',
 'Sergio Oliveira', 'Douglas Tanque', 'Fabio Abreu',
 'Brayan Riascos', 'Alex Telles ', 'Fabio Martins',
 'Haris Seferovic', 'Joao Teixeira', 'Bruno Fernandes',
 'Angel Gomes', 'Toni MartÃ\xadnez', 'Pizzi ', 'Bozhidar Kraev',
 'Sandro Lima', 'Rodrigo Pinho', 'Thiago Santana', 'Trincao ',
 'Andraz Sporar', 'Ricardo Horta', 'Bruno Duarte', 'Nuno
 Santos',
 'Domenico Berardi', 'Joao Pedro', 'Andreas Cornelius',
 'Marco Mancosu', 'Lautaro Martrinez', 'Luis Muriel',
 'Gianluca Lapadula', 'Marcus Thuram', 'Rouwen Hennings',
 'Andre Silva', 'Erling Haaland', 'Jhon Cordoba', 'Robin
 Quaison',
 'Sebastian Andersson', 'Dimitri Payet', 'Kasper Dolberg',
 'Adrien Thomasson', 'Dario Benedetto', 'Ludovic Ajorque',
 'Islam Slimani', 'Adrien Hunou', 'Denis Bouanga',
 'Sehrou Guirassy', 'Ã\x81ngel Di Maria', 'Habib Diallo',
 'Victor Osimhen', 'Dominic Calvert-Lewin', 'Kevin De Bruyne',
 'Chris Wood', 'Anthony Martial', 'Riyad Mahrez', 'Marcus
 Rashford',
 'Danny Ings', 'Richarlison ', 'Teemu Pukki', 'Tammy Abraham',
 'Thiago Galhardo', 'Paolo Guerrero', 'Pepe', 'Michael ',
 'Carlos Sanchez', 'Everaldo ', 'Artur ', 'Marcelo Cirino',
 'Yeferson Soteldo', 'Eduardo Sasha', 'Rafael Moura', 'Antony

```

None',
'Quincy Promes', 'Dusan Tadic', 'Armando Broja', 'Steven
Berghuis',
'Michael de Leeuw', 'Lois Openda', 'Danilo Nome', 'Lennart
Thy',
'Donyell Malen', 'Noni Madueke', 'Davy Klaassen',
'Oussama Tannane', 'Vaclav Cerny', 'Vangelis Pavlidis',
'Henk Veerman', 'Abdou Harroui', 'Rai Vloet', 'Lassina Traore',
'Georgios Giakoumakis', 'Alex Pozuelo', 'Kevin Molino',
'Damir Kreilach', 'Bradley Wright-Phillips', 'Nicolas Lodeiro',
'Daryl Dike', 'Cristian Pavon', 'Chris Mueller', 'Romell
Quioto',
'Gustavo Bou', 'Robert Beric', 'Ayo Akinola', 'Jeremy
Ebobisse',
'Diego Valeri', 'Youssef En-Nesyri', 'Carlos Soler',
'Cristian Tello', 'Esteban Burgos', 'Joao Felix',
'Federico Valverde', 'Kike Garcia', 'Ansu Fati', 'Roberto
Soriano',
'Gaetano Castrovilli', 'Henrikh Mkhitaryan', 'Jordan Veretout',
'Lautaro Martinez', 'Hirving Lozano', 'Lucas Alario', 'Bas
Dost',
'Dani Olmo', 'Ellyes Skhiri', 'Thomas Muller', 'Andre Hahn',
'Daniel Caligiuri', 'Matheus Cunha', 'Ludovic Blas', 'Karl
Toko',
'Burak Yilmaz', 'Ibrahima Niane', 'Boulaye Dia', 'Moise Kean',
'Ignatius Ganago', 'Irvin Cardona', 'Wissam Ben', 'Amine
Gouiri',
'Mama Balde', 'Gael Kakuta', 'James Ward-Prowse', 'Diogo Jota',
'Wilfried Zaha', 'Jack Grealish', 'Jarrod Bowen',
'Patrick Bamford', 'Danny Ings ', 'Neal Maupay', 'Ollie
Watkins',
'Luciano ', 'Vinicius ', 'Raphael Veiga', 'Luiz Adriano',
'Cleber ', 'German Cano', 'Brenner None', 'Matheus Babi',
'Alerrandro ', 'Claudinho ', 'Robson', 'Renato Kayzer',
'Donny van de Beek', 'Teun Koopmeiners', 'Cantalapiedra ',
'Bryan Linssen', 'Matavz ', 'Oussama Idrissi', 'Chidera Ejuke',
'Myron Boadu', 'Klaas-Jan Huntelaar', 'Haris Vuckic',
'Gyrano Kerk', 'Denzel Dumfries', 'Cyriel Dessers ', 'Cody
Gakpo'],
dtype=object)

```

```

check = df['Player Names'].nunique()    #Gives the number of unique
items in the colums
check

```

Removing the null values

```
link = 'https://raw.githubusercontent.com/AshishJangra27/Data-Analysis-with-Python-GFG/main/3.%20Data%20Preprocessing%20-%20Removing%20Null%20Value%20Rows/googleplaystore.csv'
```

```
data = pd.read_csv(link)
df = data.copy()
df.head()
```

	App	Category
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN
1	Coloring book moana	ART_AND_DESIGN
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN
3	Sketch - Draw & Paint	ART_AND_DESIGN
4	Pixel Draw - Number Art Coloring Book	ART_AND_DESIGN

	Reviews	Size	Installs	Type	Price	Content Rating
0	159	19M	10,000+	Free	0	Everyone
1	967	14M	500,000+	Free	0	Everyone
2	87510	8.7M	5,000,000+	Free	0	Everyone
3	215644	25M	50,000,000+	Free	0	Teen
4	967	2.8M	100,000+	Free	0	Everyone

	Genres	Last Updated	Current Ver
0	Art & Design	January 7, 2018	1.0.0
1	Art & Design;Pretend Play	January 15, 2018	2.0.0
2	Art & Design	August 1, 2018	1.2.4
3	Art & Design	June 8, 2018	Varies with device
4	Art & Design;Creativity	June 20, 2018	1.1

	Android Ver
0	4.0.3 and up
1	4.0.3 and up
2	4.0.3 and up
3	4.2 and up
4	4.4 and up

```
df.isnull().sum()
```

App	0
Category	0
Rating	1474
Reviews	0
Size	0

```
Installs      0
Type          1
Price         0
Content Rating 1
Genres        0
Last Updated  0
Current Ver    8
Android Ver    3
dtype: int64
```

```
df.dropna(inplace=True)
df.isnull().sum()
```

```
App           0
Category      0
Rating        0
Reviews       0
Size          0
Installs      0
Type          0
Price         0
Content Rating 0
Genres        0
Last Updated  0
Current Ver    0
Android Ver    0
dtype: int64
```

```
df.dropna(inplace=True, axis=0) #Drops the null values row wise
df.isnull().sum()
```

```
App           0
Category      0
Rating        0
Reviews       0
Size          0
Installs      0
Type          0
Price         0
Content Rating 0
Genres        0
Last Updated  0
Current Ver    0
Android Ver    0
dtype: int64
```

Replacing the null values

```
data.isna().sum()
```

```

App          0
Category     0
Rating      1474
Reviews      0
Size         0
Installs     0
Type         1
Price        0
Content Rating 1
Genres       0
Last Updated 0
Current Ver  8
Android Ver  3
dtype: int64

```

```

missing = round(data['Rating'].mean(),2) #This will give us the
average rating and then round it off
print(missing)
data['Rating'] = data['Rating'].fillna(missing) #This will fill the
missing values in the Ratings column
data.isna().sum()

```

4.19

```

App          0
Category     0
Rating       0
Reviews      0
Size         0
Installs     0
Type         1
Price        0
Content Rating 1
Genres       0
Last Updated 0
Current Ver  8
Android Ver  3
dtype: int64

```

Filling up categorical data

```

repeatedValues = data['Current Ver'].mode() #Mode will give us, which
value is occurring most number of times
print(repeatedValues, type(repeatedValues))
data['Current Ver'] = data['Current Ver'].fillna(str(repeatedValues))
data.isna().sum()

```

```
0    Varies with device
```

```
Name: Current Ver, dtype: object <class 'pandas.core.series.Series'>
```

```

App          0
Category     0
Rating       0
Reviews      0
Size         0
Installs     0
Type         1
Price        0
Content Rating 1
Genres       0
Last Updated 0
Current Ver  0
Android Ver  3
dtype: int64

```

Sample function

```
data.sample() #This will give a random row all the time
```

	App	Category	Rating	Reviews
Size \				
9923	CALIOPE EU: Air Quality	HEALTH_AND_FITNESS	3.9	21
2.2M				

	Installs	Type	Price	Content Rating	Genres	Last Updated \
9923	1,000+	Free	0	Everyone	Health & Fitness	October 30, 2015

	Current Ver	Android Ver
9923	1.1.2	4.0 and up

Convert the dataframe into a csv file

```

data = pd.DataFrame({
    "one": pd.Series([1, 2, 3, 4]),
    "two": pd.Series([10, 20, 30, 40]),
    "three": pd.Series([100, 200, 300, 400]),
    "four": pd.Series([1000, 2000, 3000, 4000]),
})
data.to_csv("Test File", index=False) #Converts the dataframe to csv and removes the original index

```

Pandas profile

- Gives the insight about the data with the minimum code possible